

Disproof of Conjecture 00000009675

A two-generator obstruction to the totient clause

gaochengzhecpu

Claim addressed. The conjecture concerns the two numbers $\Gamma(1/N)$ and $\Gamma(2/N)$. Its last clause asserts that, when the denominator has more than two distinct prime factors, their transcendence degree is $\varphi(N)/2$. The degree of these two numbers means the transcendence degree over $\mathbb{Q}$ of the field they generate. This clause is impossible already at $N = 105$.

Proposition 1. *For every $a, b \in \mathbb{C}$,*

$$\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(a, b) \leq 2.$$

Consequently,

$$\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(\Gamma(1/105), \Gamma(2/105)) \neq \frac{\varphi(105)}{2}.$$

Proof. The algebra $\mathbb{Q}[a, b]$ is the image of the evaluation homomorphism

$$\mathbb{Q}[X, Y] \longrightarrow \mathbb{C}, \quad X \mapsto a, \quad Y \mapsto b.$$

Its transcendence degree is therefore at most the transcendence degree 2 of the polynomial ring. Passing to its fraction field $\mathbb{Q}(a, b)$ preserves transcendence degree. This proves the bound for arbitrary a, b , without any independence assumption.

Now $105 = 3 \cdot 5 \cdot 7$ has three distinct prime divisors and $105 \geq 7$. Moreover,

$$\varphi(105) = 105 \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{7}\right) = 48, \quad \frac{\varphi(105)}{2} = 24 > 2.$$

Apply the bound to $a = \Gamma(1/105)$ and $b = \Gamma(2/105)$. Both arguments are positive, so these are ordinary finite Gamma values. The claimed degree 24 contradicts the universal upper bound 2. $\square$

Scope. Both fractions $1/105$ and $2/105$ are reduced and have denominator 105; there is no ambiguity caused by canceling a factor of 2. This disproof addresses exactly the two Gamma values named in the source. It does not replace them by the larger collection of all Gamma values at denominator N . It also makes no assertion about the separate independence clauses at $N = 5$ or $N \geq 7$.

Lean correspondence and reproduction

The Lean theorem `two_generated_trdeg_le_two` quantifies over every function $v : \text{Fin}(2) \rightarrow \mathbb{C}$ and bounds Mathlib's actual `Algebra.trdeg` of `IntermediateField.adjoin` of its range. The proof uses the genuine evaluation homomorphism from `MvPolynomial (Fin 2) ℚ`, its surjection onto the generated algebra, and the algebraic fraction-field extension to the generated field. The tower formula supplies the equality of the two transcendence degrees.

The definitions `gammaPair` and `gammaField` use Mathlib’s actual `Complex.Gamma`. The prime-factor and totient theorems use `Nat.primeFactors` and `Nat.totient`. The final theorem negates the universal totient clause and verifies $N = 105$ meets its hypotheses. No numeric approximation of a Gamma value, transcendence assumption, or substitute dimension statistic is used.

Inside `lean/`, run `lake build`, then

```
lake env lean -DwarningAsError=true Main.lean.
```

The project uses Lean 4.19.0 and Mathlib commit

```
c44e0c8ee63ca166450922a373c7409c5d26b00b.
```

All transitive dependencies are pinned in the manifest. The supplementary `verify.py` independently enumerates the totatives and prime divisors of 105 using exact integer arithmetic. The transcendence-degree bound is proved in Lean, independently of that enumeration.

The unchanged bilingual conjecture is included as `SOURCE.md`. Build logs, source hashes, theorem-axiom output, and PDF checks are in `verification/`. Author self-review and parent-agent adversarial review are separate; no external independent review is claimed.

References

- [1] The Last Math Competition, Conjecture 00000009675 (original bilingual source).